# JavaScript Conditionals and Comparison Operators Assignment

## Section 1: Basic Boolean Operations and If Statements

**Question 1.1:** What will be the output of the following code?

```javascript
console.log(5 == 5);
console.log(7 > 3);
console.log(2 < 1);
```

**Question 1.2:** What will be the output?

```javascript
let x = 25;
let y = 15;
if (x > y) {
  console.log('x is greater');
}
else {
  console.log('y is greater');
}
```

**Question 1.3:** What will be the output?

```javascript
let score = 85;
if (score >= 90) {
  console.log('Grade A');
}
else {
  console.log('Grade B');
}
```

## Section 2: Dead Code and Unreachable Code

**Question 2.1:** What will be the output?

```javascript
if (true) {
    console.log('always executed');
}
else {
    console.log('never executed');
}
```

**Question 2.2:** What will be the output?

```javascript
if (false) {
    console.log('not reachable');
}
else {
    console.log('always executed');
}
```

**Question 2.3:** What will be the output?

```javascript
let isTestPassed = true;
if (isTestPassed) {
    console.log('test passed');
}
else {
    console.log('test failed');
}
```

## Section 3: Boolean Variables in Conditions

**Question 3.1:** What will be the output?

```javascript
let isVisible = false;
if (isVisible) {
    console.log('element is visible');
}
else {
    console.log('element is hidden');
}
```

**Question 3.2:** What will be the output?

javascript

```javascript
let isLoggedIn = true;
let hasPermission = false;
if (isLoggedIn) {
    console.log('user logged in');
}
if (hasPermission) {
    console.log('access granted');
}
else {
    console.log('access denied');
}
```

## Section 4: Loose Equality (==) vs Strict Equality (===)

**Question 4.1:** What will be the output?

javascript

```javascript
let num1 = 5;
let num2 = "5";
console.log(num1 == num2);
console.log(num1 === num2);
```

**Question 4.2:** What will be the output?

javascript

```javascript
let value1 = 0;
let value2 = "0";
console.log(value1 == value2);
console.log(value1 === value2);
```

**Question 4.3:** What will be the output?

javascript

```javascript
console.log(true == 1);
console.log(true === 1);
console.log(false == 0);
console.log(false === 0);
```

## Section 5: Special Cases with Equality

**Question 5.1:** What will be the output?

```javascript
console.log("" == 0);
console.log("" === 0);
console.log([] == 0);
console.log([] === 0);
```

**Question 5.2:** What will be the output of this special case?

```javascript
console.log(null == undefined);
console.log(null === undefined);
```

**Question 5.3:** What will be the output?

```javascript
console.log(typeof null);
console.log(typeof undefined);
```

## Section 6: Conditionals with Special Values

**Question 6.1:** What will be the output?

```javascript
if (null === undefined) {
  console.log('they are strictly equal');
}
else {
  console.log('they are not strictly equal');
}
```

**Question 6.2:** What will be the output?

```javascript
if (null == undefined) {
  console.log('they are loosely equal');
}
else {
  console.log('they are not loosely equal');
}
```

**Question 6.3:** What will be the output?

javascript

```javascript
let data = null;
if (data === null) {
    console.log('data is null');
}
else {
    console.log('data has value');
}
```

## Section 7: Multiple If Statements vs If-Else Chain

**Question 7.1:** What will be the output of this code with multiple separate if statements?

javascript

```javascript
let browser = 'firefox';

if (browser === 'chrome') {
    console.log('launch chrome');
}
if (browser === 'firefox') {
    console.log('launch firefox');
}
if (browser === 'edge') {
    console.log('launch edge');
}
else {
    console.log('unsupported browser');
}
```

**Question 7.2:** What will be the output of this code with if-else chain?

javascript

```
let browser = 'firefox';

if (browser === 'chrome') {
  console.log('launch chrome');
}
else if (browser === 'firefox') {
  console.log('launch firefox');
}
else if (browser === 'edge') {
  console.log('launch edge');
}
else {
  console.log('unsupported browser');
}
```

**Question 7.3:** What will be the output?

javascript

```
let environment = 'staging';

if (environment === 'dev') {
  console.log('development environment');
}
if (environment === 'staging') {
  console.log('staging environment');
}
if (environment === 'prod') {
  console.log('production environment');
}
else {
  console.log('unknown environment');
}
```

## Section 8: Complex Conditional Scenarios

**Question 8.1:** What will be the output?

```javascript
let testStatus = 'pass';
let executionTime = 5;

if (testStatus === 'pass') {
    console.log('test passed');
}
else if (testStatus === 'fail') {
    console.log('test failed');
}
else {
    console.log('test skipped');
}

if (executionTime > 10) {
    console.log('slow execution');
}
else {
    console.log('fast execution');
}
```

**Question 8.2:** What will be the output?

```javascript
let userName = "";
let userAge = 0;

if (userName === "") {
    console.log('empty username');
}
if (userAge === 0) {
    console.log('zero age');
}
if (userName == userAge) {
    console.log('username equals age');
}
if (userName === userAge) {
    console.log('username strictly equals age');
}
```

## Section 9: Performance and Logic Issues

**Question 9.1:** What will be the output and identify the logic issue?

```javascript
let device = 'mobile';

if (device === 'desktop') {
  console.log('desktop view');
}
if (device === 'mobile') {
  console.log('mobile view');
}
if (device === 'tablet') {
  console.log('tablet view');
}
else {
  console.log('unknown device');
}
```

**Question 9.2:** What will be the output of the corrected version?

```javascript
let device = 'mobile';

if (device === 'desktop') {
  console.log('desktop view');
}
else if (device === 'mobile') {
  console.log('mobile view');
}
else if (device === 'tablet') {
  console.log('tablet view');
}
else {
  console.log('unknown device');
}
```